3. Express the rational number $\frac{1}{33}$ in recurring decimal form by using the recurring decimal expansion of $\frac{1}{11}$. Hence write $\frac{71}{33}$ in recurring decimal form.

4. Express the following decimal expression into rational numbers.

(i) $0.\overline{24}$ (ii) $2.\overline{327}$ (iii) -5.132

(iv) $3.1\overline{7}$ (v) $17.2\overline{15}$ (vi) $-21.213\overline{7}$

5. Without actual division, find which of the following rational numbers have terminating decimal expansion.

(i) $\frac{7}{128}$ (ii) $\frac{21}{15}$ (iii) $4\frac{9}{35}$ (iv) $\frac{219}{2200}$

2.3.6 Decimal Representation to Identify Irrational Numbers

It can be shown that irrational numbers, when expressed as decimal numbers, do not terminate, nor do they repeat. For example, the decimal representation of the number π starts with 3.14159265358979..., but no finite number of digits can represent π exactly, nor does it repeat.

Consider the following decimal expansions:

(i) 0.10110011100011111… (ii) 3.012012120121212…

(iii) 12.230223300222333000… (in) $\sqrt{2} = 1.4142135624\ldots$

Are the above numbers terminating (or) recurring and non- terminating? No… They are neither terminating, nor non–terminating and recurring. Hence they are not rational numbers. They cannot be written in the form of $\frac{p}{q}$,where p, q ,$\in \mathbb{Z}$ and $q \neq 0$. They are irrational numbers.

A number having non- terminating and non- recurring decimal expansion is an irrational number.

Example 2.8

Find the decimal expansion of $\sqrt{3}$.

Solution

	1.7320508…
1	3.00,00,00,00,00,…
	1
27	200
	189
343	1100
	1029
3462	7100
	6924
346405	1760000
	1732025
34641008	279750000
	277128064
	2621936

We often write $\sqrt{2} = 1.414$, $\sqrt{3} = 1.732$, $\pi = 3.14$ etc. These are only approximate values and not exact values. In the case of the irrational number π, we take frequently $\frac{22}{7}$ (which gives the value $3.\overline{142857}$) to be its correct value but in reality these are only approximations. This is because, the decimal expansion of an irrational number is non-terminating and non-recurring. None of them gives an exact value!